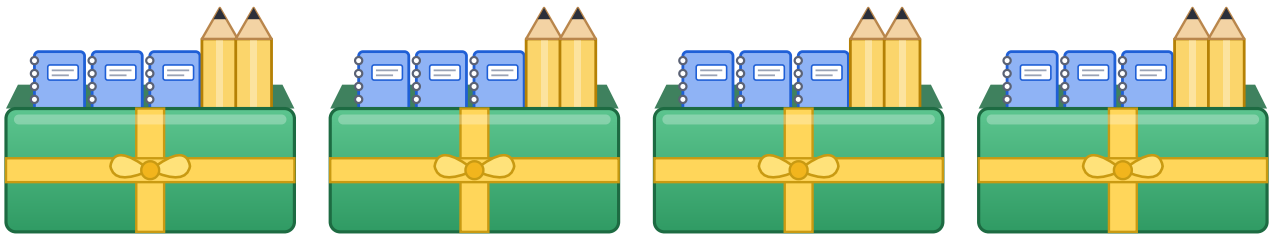
Chiziqli ifodalarda qavslarni ochish

1-QISM

Qiziqtiramiz

SOVG'A QUTILARI

Maktab olimpiadasi g'oliblari uchun 4 ta bir xil sovg'a qutisi tayyorlandi. Har bir qutiga 3 ta daftar va 2 ta qalam solindi.



1-quti 2-quti 3-quti 4-quti

1-savol. Hammasi bo'lib nechta daftar va nechta qalam kerak bo'ldi?

2-savol. Bitta daftar d so'm, bitta qalam q so'm turadi. Hamma sovg'alarning narxini ikki xil ifoda bilan yozing: avval bitta qutining narxini topib; keyin hamma daftar va hamma qalamni alohida sanab.



Bitta qutini to'rt marta olsak ham, daftarlarni va qalamlarni alohida sanasak ham, pul bir xil chiqishi kerak. Ikki ifodani qanday bog'lash mumkin?

ESLAB OLAYLIK

- O'xshash hadlar ixchamlanadi: $5y - 6y = -y$, $3a + 2 - a = 2a + 2$.
- Ishoralar: $(-4) \cdot 3 = -12$, $(-2) \cdot (-5) = 10$.
- Maxrajlarni tenglashtiramiz: $\frac{1}{4} + \frac{1}{6} = \frac{3}{12} + \frac{2}{12} = \frac{5}{12}$; $\frac{2x}{5} = \frac{2}{5}x$.

2-QISM

Sinab ko'ramiz

1-TOPSHIRIQ · BIR XIL GURUHLAR

Yashil tugma — x, sariq — 1, qizil — $-x$ va -1. Rasmda $3(2x)$: $2x$ ning 3 ta guruhi.



a) Nechta x tugma bor? $3(2x) = ?$

b) $x + 2$ ning 3 ta guruhida nechta x va nechta 1 bo'ladi?

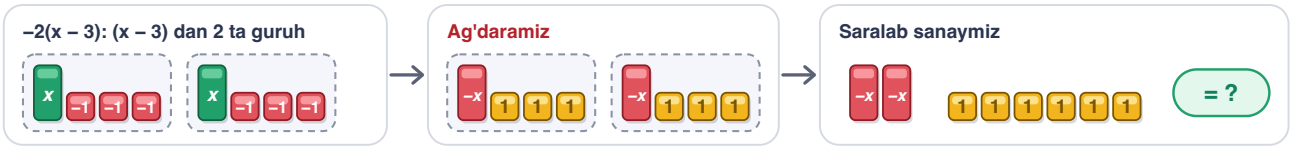
2-TOPSHIRIQ · JADVAL

Jadvalni to'ldiring. Qaysi ikkita ifoda x ning har qanday qiymatida teng chiqadi?

x	$5(x-2)$	$5x-10$	$5x-2$
1	?	?	?
3	?	?	?
6	?	?	?

3-TOPSHIRIQ · GURUHLARNI AG'DARAMIZ

Qavs oldida manfiy son bo'lsa, guruhlarni olib, hamma tugmani ag'daramiz (har bir tugma qarama-qarshisiga aylanadi). Rasmda $-2(x-3)$.



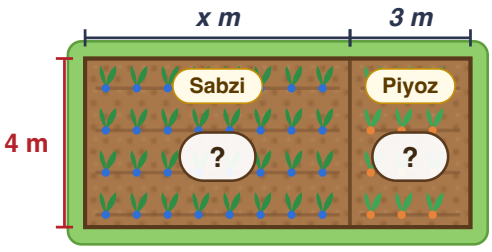
Ag'darilgandan keyin qanday tugmalar qoldi? $-2(x-3)$ nimaga teng?

4-TOPSHIRIQ · TOMORQA

Tomorqa to'g'ri to'rtburchak shaklida, eni 4 m. U ikki pushtaga bo'lingan: sabzi pushtasining bo'yi x m, piyoz pushtasiniki 3 m.

a) Har bir pushtaning yuzini toping.

b) Butun tomorqaning yuzini ikki xil ifoda bilan yozing.



3-QISM

Tushunamiz

$3(x-2)$ ko'rinishidagi ifodada qavslarni ochib, uni hadlarining yig'indisi ko'rinishida yozish mumkin. Buni algebraik tugmalar bilan ko'rsatamiz.

USUL · ALGEBRAIK TUGMALAR BILAN QAVSLARNI OCHISH

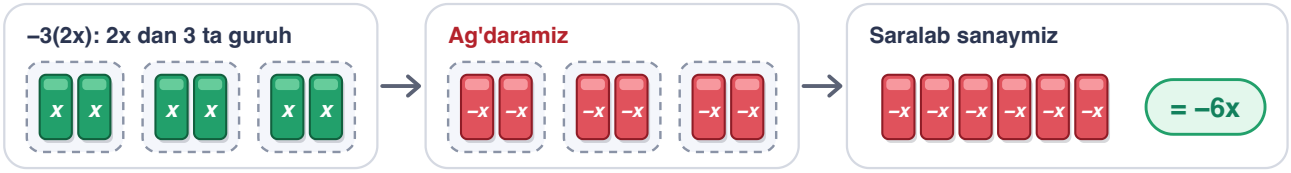
1. Qavs oldidagi son musbat bo'lsa: qavs ichidagi ifodadan shuncha guruh olamiz, so'ng tugmalarni ajratib sanaymiz.
2. Qavs oldidagi son manfiy bo'lsa: shuncha guruh olamiz (minus belgisiz), so'ng hamma tugmalarni ag'daramiz. «–» ishorasi guruhlardagi barcha tugmalarni qarama-qarshisiga aylantirishni (ishorasini o'zgartirishni) anglatadi.

1-MISOL · MUSBAT KO'PAYTUVCHI



$$2(4y-1)=8y-2$$

2-MISOL · MANFIY KO'PAYTUVCHI



$$-3(2x)=-(6x)=-6x$$

QOIDA · TAQSIMOT QONUNI

Qavslarni ochish jarayonida ko'paytirishning taqsimot qonunidan foydalaniladi:

$$a(x+y) = ax + ay$$

$a(x+y)$ ifoda $ax+ay$ ko'rinishida **yoyilgan** yoki $ax+ay$ ifoda $a(x+y)$ ning **yoyilgan shakli** deymiz.

Taqsimot qonunini quyidagicha umumlashtirish va qo'llash mumkin:

- $a(x+y) = ax + ay$ — ko'paytirish qo'shishga nisbatan taqsimlanishi mumkin.
- $a(x-y) = ax + a(-y) = ax - ay$ — ko'paytirish ayirishga nisbatan taqsimlanishi mumkin.
- $a(x+y+z) = ax + ay + az$ — ko'paytirish bir nechta hadlarni qo'shishga nisbatan taqsimlanishi mumkin.

E'tibor bering: $2(5) = 2 \cdot 5$, $2(3x) = 2 \cdot 3x$. Ko'paytuvchi qavsdan keyin kelsa ham qoida o'zgarmaydi: $(x+y)a = xa + ya$.

SOVG'ALAR VA TOMORQA

Sovg'alar: 12 ta daftar, 8 ta qalam; $4(3d+2q) = 12d + 8q$. Tomorqa: $4(x+3) = 4x + 12$.

3-MISOL · UCHTA HAD

5 ni har bir hadga ko'paytiramiz:

$$5(3x - 2y + 4)$$
$$= 5(3x) - 5(2y) + 5(4)$$
$$= 15x - 10y + 20$$

4-MISOL · KO'PAYTUVCHI O'NGDA

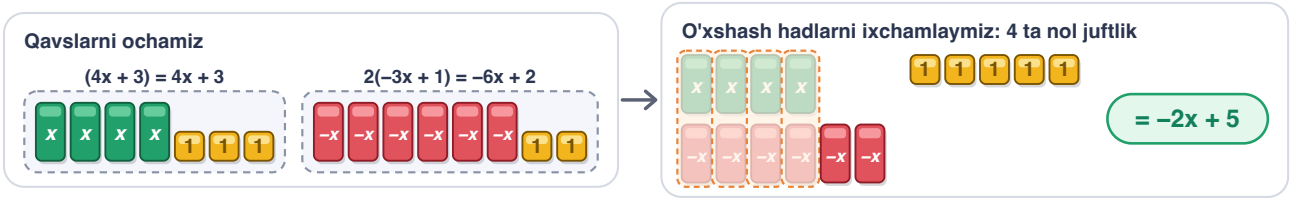
$-4k$ ni har bir hadga ko'paytiramiz:

$$(2m - 3n)(-4k)$$
$$= (2m)(-4k) + (-3n)(-4k)$$
$$= -8mk + 12nk$$

QOIDA · QAVSLARNI OCHISH VA SODDALASHTIRISH

Ifodani soddalashtirish avval qavslarni ochish, so'ng o'xshash hadlarni ixchamlash orqali amalga oshiriladi.

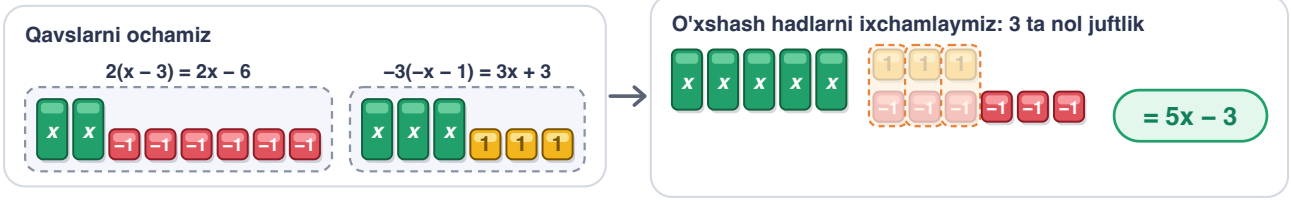
5-MISOL · ISHORALAR O'ZGARMAYDI



$(4x+3)+2(-3x+1)=4x+3-6x+2$ — qavs oldida musbat son, ishoralar o'zgarmaydi

$=4x-6x+3+2=-2x+5$

6-MISOL · ISHORALAR O'ZGARADI



$2(x-3)-3(-x-1)=2x-6+3x+3$ — ikkinchi qavs oldida manfiy son, ishoralar o'zgaradi

$=2x+3x-6+3=5x-3$

TA'RIF · TENG KUCHLI IFODALAR

Ikkita ifoda soddalashtirilganda bir xil ifoda kelib chiqsa, ular **teng kuchli ifodalar** deyiladi.

7-MISOL · TENG KUCHLIMI?

$3(2x-1)-2(x-4)$

$=6x-3-2x+8=4x+5$

$2(3x+1)-(2x-3)$

$=6x+2-2x+3=4x+5$

Ikkalasi ham $4x+5$ ga keldi. Demak, bu ikki ifoda va $4x+5$ — uchalasi teng kuchli.

8-MISOL · ICHMA-ICH QAVSLAR

Eng oldin ichki qavslarni oching.

$4[5-3(x-2)]$

$=4[5-3x+6]$ — ichki qavs

$=4(11-3x)=44-12x$

QOIDA · ODDIY KASRLAR QATNASHGAN IFODALAR

Oddiy kasrlar qatnashgan ifodalarni ularning hadlarini maxraji bir xil bo'lgan kasrlarga keltirish orqali soddalashtirish mumkin.

1. Maxrajlarning EKUK ini topib, maxrajlarni tenglashtiramiz.
2. Yagona oddiy kasr ko'rinishida yozamiz (har bir suratni qavsga olib).
3. Taqsimot qonunini qo'llaymiz.
4. O'xshash hadlarni ixchamlaymiz.

9-MISOL · KASRLAR YIG'INDISI

$\frac{2x+1}{3}+\frac{x-2}{4}$, EKUK(3; 4) = 12.

$=\frac{4(2x+1)+3(x-2)}{12}$

$=\frac{8x+4+3x-6}{12}=\frac{11x-2}{12}$

10-MISOL · KASRLAR AYIRMASI

$\frac{3x}{4}-\frac{2(x-3)}{5}$, EKUK(4; 5) = 20.

$=\frac{15x-8(x-3)}{20}$ — minus butun suratga tegishli, suratni qavsga olamiz

$=\frac{15x-8x+24}{20}=\frac{7x+24}{20}$

4-QISM

Qo'llaymiz

1-MISOL · MANFIY VA KASR KO'PAYTUVCHI

Qavslarni oching: a) $(-5)(-2x+3y)$ b) $\frac{3}{4}(8a-12b+4)$

a) -5 ni har bir hadga ko'paytiramiz: $(-5)(-2x)+(-5)(3y)=10x-15y$.

b) $\frac{3}{4}$ ni har bir hadga: $\frac{3}{4}\cdot 8a-\frac{3}{4}\cdot 12b+\frac{3}{4}\cdot 4=6a-9b+3$.

2-MISOL · HARFLI KO'PAYTUVCHI

Soddalashtiring: $y(2x-3)+3y(x+1)$

1. Qavslarni ochamiz: $2xy-3y+3xy+3y$.
2. O'xshash hadlar: $2xy+3xy=5xy$, $-3y+3y=0$. Javob: $5xy$.

3-MISOL · IKKI HARFLI IFODA

Soddalashtiring: $-3(2a-4b)+5(a-3b)$

1. -3 ni birinchi qavsdagi har bir hadga: $-6a+12b$ ($(-)\cdot(-)=(+)$).
2. 5 ni ikkinchi qavsdagi har bir hadga: $5a-15b$.
3. $-6a+5a+12b-15b=-a-3b$.

4-MISOL · ICHMA-ICH QAVSLAR

Soddalashtiring: $3y-2[y-(4-5y)]$

1. Ichki qavs oldida minus — ishoralar o'zgaradi: $y-4+5y=6y-4$.
2. $3y-2(6y-4)=3y-12y+8$ ($-2\cdot(-4)=8$).
3. $=-9y+8$.

5-MISOL · UCHTA KASR

Bitta qisqarmas kasr ko'rinishida yozing: $\dfrac{x+2}{3}-\dfrac{3x-1}{4}+\dfrac{x}{6}$

1. EKUK(3; 4; 6) = 12: $\dfrac{4(x+2)-3(3x-1)+2x}{12}$.
2. Qavslarni ochamiz: $\dfrac{4x+8-9x+3+2x}{12}$.
3. $=\dfrac{-3x+11}{12}$. Surat va maxrajning umumiy bo'luvchisi yo'q — kasr qisqarmas.

6-MISOL · KONSERT ZALI

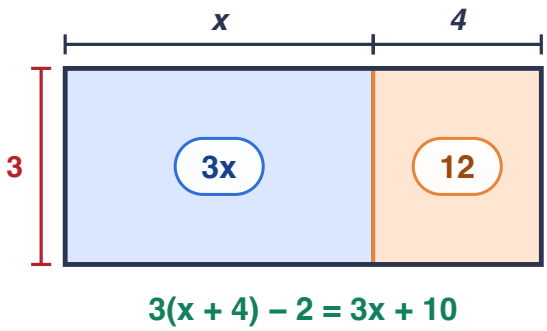
Konsert zalining chap qismida 8 qator, har qatorda $(x+4)$ ta o'rindiq; o'ng qismida 6 qator, har qatorda $(x-2)$ ta o'rindiq bor. a) Jami o'rindiqlar sonini yoyilgan ifoda shaklida yozing. b) $x=15$ bo'lsa, zalda nechta o'rindiq bor?

a $8(x+4)+6(x-2)=8x+32+6x-12=14x+20$.

b $14\cdot 15+20=210+20=230$ ta. Tekshiramiz: $8\cdot 19+6\cdot 13=152+78=230$.

7-MISOL · BO'YALADIGAN DEVOR

Devorning balandligi 3 m, uzunligi $(x+4)$ m. Devordagi derazaning yuzi 2 m². Bo'yaladigan qism yuzini x orqali yozing.



1 Butun devor: $3(x+4)=3x+12$ m².

2 Derazani ayiramiz: $3x+12-2=3x+10$ m².



Javobni tekshirish oson: x o'rniga biror son qo'yaman — berilgan ifoda ham, javob ham bir xil son chiqishi kerak.

DIQQAT!

1. Qavsdagi har bir hadga ko'paytiriladi: ✗ $3(2x+5)=6x+5$ ✓ $3(2x+5)=6x+15$
2. Manfiy son ishorani o'zgartiradi: ✗ $-2(x-4)=-2x-8$ ✓ $-2(x-4)=-2x+8$
3. Qavs oldidagi minus — bu -1 ga ko'paytirish: ✗ $5x-(2x-3)=3x-3$ ✓ $5x-2x+3=3x+3$
4. Kasr oldidagi minus butun suratga tegishli: ✗ $\frac{x}{2}-\frac{x-1}{3}=\frac{3x-2x-1}{6}$ ✓ $\frac{3x-2(x-1)}{6}=\frac{x+2}{6}$
5. Avval ko'paytirish, keyin ayirish: ✗ $4-3(x-2)=1(x-2)$ ✓ $4-3x+6=10-3x$

Mashqlar

1. Qavslarni oching.
   a) $4(3y)$ b) $5(-2a)$ c) $-6(4m)$
   d) $-3(-7n)$ e) $-(8x)$ f) $-1(-9k)$

2. Qavslarni oching.
   a) $3(x+5)$ b) $7(2a-3)$ c) $-4(3y+2)$
   d) $-5(-2m-6)$ e) $(6p-1)\cdot 3$ f) $(-2)(4k-7)$

3. Qavslarni oching.
   a) $2(4a+3b-5)$ b) $-3(2x-y+4z)$ c) $m(5n-2)$
   d) $-a(3b+4c-1)$ e) $(2u-3v+6)(-4w)$

4. Qavslarni oching.
   a) $\frac{1}{2}(8x-6)$ b) $\frac{3}{5}(10a+15b-5)$
   c) $-\frac{2}{3}(9m-12n+3)$ d) $(6x-4y+10)\cdot\left(-\frac{1}{2}\right)$

5. Rasmdagi tugmalar qaysi ifodaga mos? Qavslarni ochib, natijani yozing.



6. Qavslarni oching va soddalashtiring.
   a) $3(x+4)+2(x-1)$ b) $5(2a-1)-3(a+2)$
   c) $-2(3y-4)+4(y-3)$ d) $4(m-2n)-3(2m-n)$
   e) $6(2k+1)-(5k-4)$ f) $-(x-7)-2(3-x)$

7. Soddalashtiring.
   a) $7x-2(3x-5)$ b) $-4(2a+3)+9a$
   c) $3(2p-q+4)-2(p-3q)$ d) $a(2b-5)+3a(1-b)$
   e) $5(a-2b)+(3a+b-2)(-2)$

8. Qavslarni oching va soddalashtiring.
   a) $3[2x+(x-4)]$ b) $8a-[3a+(a-5)]$
   c) $2[3y-4(y-2)]$ d) $4m-3[m-2(3-m)]$
   e) $2(p+3q)-[4p-(q-p)]$ f) $\frac{1}{2}[10x-4(x+3)]$

9. Bitta qisqarmas oddiy kasr ko'rinishida yozing.
   a) $\frac{x}{4}+\frac{x}{6}$ b) $\frac{2a}{3}-\frac{a}{5}$ c) $\frac{x+3}{2}+\frac{x-1}{5}$
   d) $\frac{3y-2}{4}-\frac{y+1}{6}$ e) $\frac{5n}{6}-\frac{3(n-2)}{4}$
   f) $\frac{x+1}{3}+\frac{2x-1}{4}-\frac{x-2}{6}$

10. Qaysi ifodalar teng kuchli? Qaysi biri ularga teng kuchli emas?
    A) $4(2x-3)-3(x-5)$ B) $5x+3$ C) $2(3x+4)-(x+5)$
    D) $2(3x+1)-(x-2)$ E) $-(3-5x)+6$

11. $\square$ o'rniga qaysi son yoziladi?
    a) $3(2x+\square)=6x+12$ b) $\square(4a-1)=-8a+2$
    c) $-5(\square x-2y)=15x+10y$ d) $2(x+3)+\square(x-1)=6x+2$

12. Avval soddalashtiring, so'ng qiymatini toping.
    a) $3(2a-5)-4(a-6),\ a=7{,}5$ b) $-2(x+4)+5(x-1),\ x=-3$

13. **Xatoni toping.** Sardor shunday yechdi: $-2(3x-4)+x=-6x-8+x=-5x-8$. Qayerda xato qildi? To'g'ri javob qanday?

14. Sinf muzeyga bordi. O'qituvchi chiptasi k so'm, o'quvchi chiptasi undan 15 000 so'm arzon. 1 ta o'qituvchi va 24 ta o'quvchi chiptasi olindi, yana muzey gidiga 50 000 so'm to'landi.
    a) Jami xarajatni k orqali yozing va soddalashtiring.
    b) $k=30\,000$ bo'lsa, jami qancha pul ketgan?

15. Polizda 4 ta bir xil egat bor. Har bir egatga $(3n+2)$ tup pomidor va $(n-1)$ tup qalampir ekildi.
    a) Jami ko'chatlar sonini ikki usulda yozing: $4(3n+2)+4(n-1)$ va $4[(3n+2)+(n-1)]$. Ikkalasini soddalashtiring.
    b) $n=10$ bo'lsa, nechta ko'chat ekilgan?

16. Liftga har biri $(2m+5)$ kg bo'lgan 6 ta quti va har biri $(m-3)$ kg bo'lgan 4 ta qop yuklandi.
    a) Yukning jami massasini yoyilgan ifoda shaklida yozing.
    b) $m=20$ bo'lsa, yuk necha kg?

17. Hovuzda $(8x+20)$ litr suv bor edi. Birinchi soatda suvning $\frac{1}{4}$ qismi, ikkinchi soatda yana 6 litr chiqarildi.
    a) Hovuzda qancha suv qolganini x orqali yozing va soddalashtiring.
    b) $x=5$ bo'lsa, necha litr suv qolgan?

18. To'g'ri to'rtburchakning bo'yi $(3x+2)$ cm, eni $(x-1)$ cm. Uning perimetrini x orqali yozing. $x=4$ bo'lsa, perimetr nechaga teng?

5-QISM

Tekshiramiz

1. Qavslarni oching.
    a) $6(2x-3)$ b) $-4(5a+1)$ c) $(3m-2n)(-5)$
    d) $-2(-4p+3q-1)$ e) $\frac{2}{3}(9x-6)$

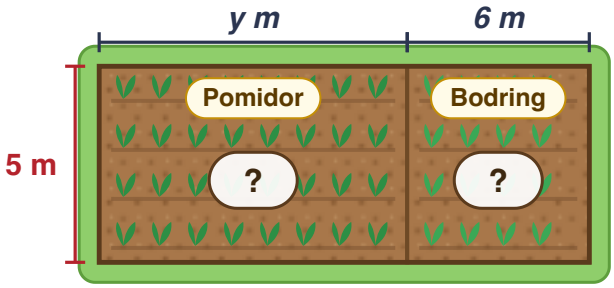
2. Qavslarni oching va soddalashtiring.
    a) $4(2y+3)-5(y-1)$ b) $-3(a-2b)+2(4a-b)$
    c) $9k-(4k-7)$ d) $x(3-y)+2x(y+1)$

3. Soddalashtiring: a) $2[5x - 3(x + 2)]$ b) $7a - [3a + 2(a - 4)]$

4. Bitta qisqarmas oddiy kasr ko'rinishida yozing: a) $\frac{x+4}{3} + \frac{x-2}{2}$ b) $\frac{5y-1}{6} - \frac{2(y+1)}{9}$

5. Qaysi ikkitasi teng kuchli: $3(x - 2) + 4$; $2(2x - 1) - x$; $5(x - 1) - 2(x - 2)$?

6. Rasmdagi tomorqaning umumiy yuzini ikki xil ifoda bilan yozing.



7. Telefon tarifi uchun har oy $(3t + 10)$ ming so'm to'lanadi. Aziz 12 oy to'ladi. 2 oyda chegirma bo'ldi: o'sha oylarning har birida t ming so'm kam to'ladi. Jami to'lovni t orqali yozing va soddalashtiring. $t = 5$ bo'lsa, qancha to'lagan?

8. **Xatoni toping.** Nodira: «$\frac{5x}{6} - \frac{x+3}{4} = \frac{10x - 3x + 9}{12} = \frac{7x+9}{12}$». To'g'ri javob qanday?

9. $a(2x - 3) - 4x$ ifodada a ning qanday qiymatida x qatnashmaydi? Shunda ifoda nimaga teng bo'ladi?

- [ ] Men taqsimot qonuni yordamida qavslarni ocha olaman — qavs oldidagi son musbat ham, manfiy ham bo'lganda.
- [ ] Men avval qavslarni ochib, so'ng o'xshash hadlarni ixchamlab ifodani soddalashtira olaman; ichma-ich qavslarni ichkaridan boshlab ochaman.
- [ ] Men ikki ifoda teng kuchli ekanini soddalashtirib tekshira olaman.
- [ ] Men oddiy kasrlar qatnashgan ifodani bitta qisqarmas kasr ko'rinishida yoza olaman.

Javoblar

4-qism · Mashqlar

1-masala

a) $12y$. b) $-10a$. c) $-24m$. d) $21n$. e) $-8x$. f) $9k$.

2-masala

a) $3x + 15$. b) $14a - 21$. c) $-12y - 8$. d) $10m + 30$. e) $18p - 3$. f) $-8k + 14$.

3-masala

a) $8a + 6b - 10$. b) $-6x + 3y - 12z$. c) $5mn - 2m$. d) $-3ab - 4ac + a$. e) $-8uw + 12vw - 24w$.

4-masala

a) $4x - 3$. b) $6a + 9b - 3$. c) $-6m + 8n - 2$. d) $-3x + 2y - 5$.

5-masala

$3x - 2$ dan 2 ta guruh, oldida minus: $-2(3x - 2)$. Ag'darsak — 6 ta $-x$ va 4 ta 1: $-6x + 4$.

6-masala

a) $3x + 12 + 2x - 2 = 5x + 10$. b) $10a - 5 - 3a - 6 = 7a - 11$. c) $-6y + 8 + 4y - 12 = -2y - 4$. d) $4m - 8n - 6m + 3n = -2m - 5n$. e) $12k + 6 - 5k + 4 = 7k + 10$. f) $-x + 7 - 6 + 2x = x + 1$.

7-masala

a) $7x - 6x + 10 = x + 10$. b) $-8a - 12 + 9a = a - 12$. c) $6p - 3q + 12 - 2p + 6q = 4p + 3q + 12$. d) $2ab - 5a + 3a - 3ab = -ab - 2a$. e) $5a - 10b - 6a - 2b + 4 = -a - 12b + 4$.

8-masala

a) $3(3x-4)=9x-12$. b) $8a-(4a-5)=8a-4a+5=4a+5$. c) $2(3y-4y+8)=2(-y+8)=-2y+16$. d) $m-6+2m=3m-6$; $4m-3(3m-6)=4m-9m+18=-5m+18$. e) Kvadrat qavs: $4p-q+p=5p-q$; $2p+6q-(5p-q)=2p+6q-5p+q=-3p+7q$. f) $\frac{1}{2}(10x-4x-12)=\frac{1}{2}(6x-12)=3x-6$.

9-masala

a) $\frac{3x+2x}{12}=\frac{5x}{12}$. b) $\frac{10a-3a}{15}=\frac{7a}{15}$. c) $\frac{5(x+3)+2(x-1)}{10}=\frac{7x+13}{10}$. d) $\frac{3(3y-2)-2(y+1)}{12}=\frac{7y-8}{12}$. e) $\frac{10n-9(n-2)}{12}=\frac{n+18}{12}$. f) $\frac{4(x+1)+3(2x-1)-2(x-2)}{12}=\frac{8x+5}{12}$.

10-masala

A) $8x-12-3x+15=5x+3$. C) $6x+8-x-5=5x+3$. D) $6x+2-x+2=5x+4$. E) $-3+5x+6=5x+3$. A, B, C, E — teng kuchli; D ularga teng kuchli emas.

11-masala

a) $3\cdot\square=12$, $\square=4$. b) $\square\cdot 4a=-8a$, $\square=-2$ (tekshiramiz: $-2\cdot(-1)=2$). c) $-5\cdot\square x=15x$, $\square=-3$ ($-5\cdot(-2y)=10y$). d) $2x+6+\square x-\square=6x+2$: $\square=4$ ($2x+4x=6x$, $6-4=2$).

12-masala

a) $6a-15-4a+24=2a+9$; $2\cdot 7{,}5+9=24$. b) $-2x-8+5x-5=3x-13$; $3\cdot(-3)-13=-22$.

13-masala

$-2\cdot(-4)=8$ — musbat. Sardor ishorani o'zgartirmagan. To'g'risi: $-6x+8+x=-5x+8$.

14-masala

a) $k+24(k-15\,000)+50\,000=k+24k-360\,000+50\,000=25k-310\,000$. b) $25\cdot 30\,000-310\,000=440\,000$ so'm.

15-masala

a) $12n + 8 + 4n - 4 = 16n + 4$; $4(4n + 1) = 16n + 4$ — bir xil. b) $16 \cdot 10 + 4 = 164$ ta.

16-masala

a) $6(2m + 5) + 4(m - 3) = 12m + 30 + 4m - 12 = 16m + 18$. b) $16 \cdot 20 + 18 = 338$ kg.

17-masala

a) $(8x + 20) - \frac{1}{4}(8x + 20) - 6 = 8x + 20 - 2x - 5 - 6 = 6x + 9$. b) $6 \cdot 5 + 9 = 39$ litr.

18-masala

$2(3x + 2) + 2(x - 1) = 6x + 4 + 2x - 2 = 8x + 2$. $x = 4$: $8 \cdot 4 + 2 = 34$ cm.

5-qism · Tekshiramiz

1-masala

a) $12x - 18$. b) $-20a - 4$. c) $-15m + 10n$. d) $8p - 6q + 2$. e) $6x - 4$.

2-masala

a) $8y + 12 - 5y + 5 = 3y + 17$. b) $-3a + 6b + 8a - 2b = 5a + 4b$. c) $9k - 4k + 7 = 5k + 7$. d) $3x - xy + 2xy + 2x = 5x + xy$.

3-masala

a) $2(5x - 3x - 6) = 2(2x - 6) = 4x - 12$. b) $7a - (3a + 2a - 8) = 7a - 5a + 8 = 2a + 8$.

4-masala

a) $\frac{2(x + 4) + 3(x - 2)}{6} = \frac{5x + 2}{6}$. b) $\frac{3(5y - 1) - 4(y + 1)}{18} = \frac{11y - 7}{18}$.

5-masala

$3x - 6 + 4 = 3x - 2$; $4x - 2 - x = 3x - 2$; $5x - 5 - 2x + 4 = 3x - 1$. Birinchi ikkitasi teng kuchli.

6-masala

$5(y+6)$ va $5y+30$ (m²).

7-masala

$12(3t+10)-2t=36t+120-2t=34t+120$. $t=5$: $170+120=290$ ming so'm.

8-masala

Minus butun suratga tegishli: $\dfrac{10x-3(x+3)}{12}=\dfrac{10x-3x-9}{12}=\dfrac{7x-9}{12}$.

9-masala

$2ax-3a-4x$: x qatnashmasligi uchun $2a-4=0$, $a=2$. Shunda ifoda -6 ga teng.